Lines of Best Fit

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Line of Best Fit

What is a line of best fit?

- If a **scatter graph** suggests that there is a positive or negative **correlation**
  - a **line of best fit** can be **drawn** on the scatter graph
    - This can then be used to make **predictions**

How do I find the coordinates of the mean point?

- You need to be able to find the **mean point**, $(\bar{x}, \bar{y})$ to draw a line of best fit
  - Calculate the **mean of the *x* values** for all data points, this is $\bar{x}$
  - Calculate the **mean of the *y* values** for all data points, this is $\bar{y}$
- The line of best fit **must pass through the mean point**

How do I draw a line of best fit?

- To draw a **line of best fit**:
  - **Find** the **coordinates** of the **mean point**, $(\bar{x}, \bar{y})$
  - **Plot** the **mean point** on the graph with all of the other data values
  - Draw a **single-ruled straight line** through the mean point
    - It must extend across the **full** data set
    - **Adjust the angle** of the line to try and get roughly the **same number of data points** on either side of the line along its **whole** length
- If there is one **extreme value** (outlier) that does not fit the general pattern
  - then **ignore** this point when drawing a line of best fit

How do I use a line of best fit?

- Once the line of best fit is drawn, you can use it to **predict values**
  - E.g. to estimate *y* when *x* = 5
    - Use the line to read off the *y* value when *x* is 5
- It is best to use your line to predict values that lie **within the region** covered by the data points

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

line are **unreliable**!
    - This is called **extrapolation**

Examiner Tips and Tricks

- Placing a ruler so that it goes through the mean point and tilting it forwards and backwards can help to find the right position for the line of best fit!

Worked Example

Sophie wants to know if the price of a computer is related to the speed of the computer.
She tests 8 computers by running the same program on each, measuring how many seconds it takes to finish.
Sophie's results are shown in the table below.

Price (£)	320	300	400	650	250	380	900	700
Time (secs)	3.2	5.4	4.1	2.8	5.1	4.3	2.6	3.7

(a) Draw a scatter diagram, showing the results on the axes below.

Plot each point carefully using crosses

(b) Write down the type of correlation shown and use it to form a suitable conclusion.

The shape formed by the points goes from top left to bottom right (a negative gradient)
This is a negative correlation
As one quantity increases (price), the other decreases (time)

(c) Use a line of best fit to estimate the price of a computer that completes the task in 3.4 seconds.

Find the mean point, $(\bar{x}, \bar{y})$

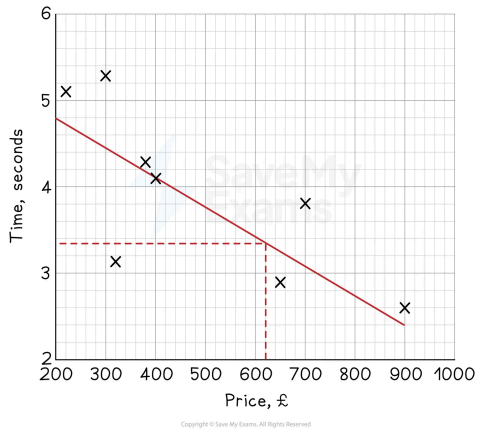
$$\bar{x} = \frac{320 + 300 + 400 + 650 + 250 + 380 + 900 + 700}{8} = 487.5$$

$$\bar{y} = \frac{3.2 + 5.4 + 4.1 + 2.8 + 5.1 + 4.3 + 2.6 + 3.7}{8} = 3.9$$

$$(\bar{x}, \bar{y}) = (487.5, 3.9)$$

Plot $(\bar{x}, \bar{y})$ on the graph
Draw a line of best fit that goes through it

Then draw a horizontal line from 3.4 seconds to the line of best fit
Draw a vertical line down to read off the price



A computer that takes 3.4 seconds to run the program should cost around £620

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